Nazar Budaiev

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Education

Ph.D. in Astronomy, University of Florida — *May 2026*
Dissertation: Extreme Star Formation in Sagittarius B2. Advisor: Adam Ginsburg

M.S. in Astronomy, University of Florida — *August 2022*

B.A. in Astronomy & Physics, Boston University — *May 2020*

Research Experience

Postdoctoral Associate — *2026 – present*
University of Florida
- Leading data reduction and analysis efforts for a JWST large program covering the Galactic Center (PID: 10678).
- Designing and maintaining a pipeline that improves astrometric accuracy and source extraction for complex fields.

Graduate Research Assistant — *2020 – 2026*
University of Florida
- Investigated star formation in the Galactic Center with JWST, ALMA, and VLA; primary focus on Sagittarius B2.
- Evaluated and re-imaged data as part of data reduction team for ACES ALMA large program, coordinating with an international team of >100 investigators across multiple institutions.
- Lead author on 3 and n-th author on 15 refereed publications; contributed to 13 accepted telescope proposals across JWST, ALMA, and VLA, with experience in proposal review.

Visiting Researcher — *Spring 2025*
University of Connecticut
- Consulted on source extraction and classification strategies for the ACES large ALMA program.
- Analyzed and characterized star-forming populations in the Galactic center using the ACES data.

Research Assistant — *2018 – 2020*
Boston University
- Characterized GBT's VEGAS receiver; modeled HI clouds in the Milky Way with IDL.

Summer Research Intern — *Summer 2019*
Green Bank Observatory
- Created a 6 GHz continuum map of the inner Galactic Plane using GBT data in IDL and Python.

Selected Talks

- *"Star formation in the Galactic center,"* **Invited seminar**. Yale University, New Haven, CT — *Oct 2025*
- *"Multiwavelength mysteries in Sgr B2,"* **Invited**. TUNA Talk, NRAO, Charlottesville, VA — *Oct 2025*
- *"Multiwavelength mysteries in Sgr B2,"* Contributed talk. Stellar Origins, Austria — *Sep 2025*
- *"Multiwavelength mysteries in Sgr B2,"* **Invited seminar**. ESO, Germany — *Sep 2025*
- *"Star formation in Sgr B2,"* Contributed talk. ALMA-IMF workshop (remote) — *Jun 2025*
- *"Star formation in Sgr B2,"* Contributed talk. STScI Symposium, Baltimore — *May 2025*
- *"Where are the stars in Sgr B2?"* **Invited colloquium**. University of Connecticut, Storrs, CT — *Apr 2025*
- *"Star formation in Sgr B2,"* Contributed talk. ACES meeting, Boston, MA — *Aug 2024*
- *"Protostellar cores in Sgr B2,"* **Invited seminar**. Kansas University (remote) — *Mar 2024*
- *"Protostellar cores in Sgr B2,"* Contributed talk. Surveying the Milky Way, Pasadena, CA — *Oct 2023*

Awards

ALMA Ambassador, Cycle 13, **$2,500** — *2026*
Various Travel Grants, **$3,200** — *2025*
JWST Cycle 3 GO 5365 — Co-PI (17.9 hrs) — *2024*
ALMA Cycle 11 — PI (6.9 hrs) — *2024*
Certificate of Outstanding Merit, UF International Center — *2023*

Symposium Talk Award, UF Astronomy *2023*
Distinguished Service & Citizenship, UF Astronomy *2022*
NRAO Student Observing Support, ALMA Cycle 8, **$34,955** *2021*
Undergraduate Research Award, BU Astronomy *2020*

Workshops

Code/Astro, Evanston, IL *2024*
19th Synthesis Imaging Workshop, Charlottesville, VA *2023*
Star Formation School, Granada, Spain *2021*
GBT Remote Observer Certification, Green Bank Observatory, WV *2021*
Single Dish School, Green Bank Observatory, WV *2021*

Teaching & Mentoring

Project Advisor *Fall 2024 – present*
UF Star Formation Group
- Advising undergraduate (A. Daley) on a YSO catalog in Sgr B2 with ALMA. Student received $1,750 UF University Scholars award. A publication is on track to be submitted in Fall 2026.

Graduate Student Instructor *2020 – 2021*
Astronomy Laboratory, UF
- Taught undergraduate lab course; redesigned outdated experiments; produced remote-learning videos.

Teaching Assistant *Spring 2020*
Stellar & Galactic Astrophysics, Boston University
- Assisted students in an IDL-heavy upper-level astrophysics course.

Teaching Assistant *2018 – 2020*
Project Accelerate, Boston University
- Led lab sections of an Advanced Placement physics course for underserved students; developed review materials for each course module.

Science Communication & Community Engagement

- **Referee**, AAS Journals *2025 – present*
- **Webinar Author**, AstroSandbox — wrote and delivered 6 astronomy webinars for Ukrainian students *Python in astronomy*, *Astronomical image processing*, *Multiwavelength astronomy*, *Radio astronomy*, *How to write a telescope proposal?*, *Imposter syndrome*. *2020 – present*
- **Competition Judge**, AstroSandbox — judged annual astronomy team competition with 50 participants from 15 regions of Ukraine. *2020 – 2023*
- **Graduate Student Representative**, Mentoring Committee, UF Astronomy Department — gathered community feedback, synthesized findings into anonymized reports, and presented recommendations to faculty; served as liaison between student body and departmental leadership *2021 – 2023*
- **Mentor**, Ukraine Global Scholars, Ukraine Achievement Fund — guided 4 high-school students through university applications; 2 admitted to top-20 US boarding schools *2020 – 2024*
- **Mentor**, UF Women in Astrophysics & Astronomy Mentorship Program *2022 – 2024*
- **Pen Pal**, Letters to a Pre-Scientist. Year-long physical mail exchange with a high-school student *2021 – 2022*

Media Coverage

- UF press release: *UF researchers help capture most detailed view of the Milky Way's core* *2026*
- NASA press release: *NASA's Webb Explores Largest Star-Forming Cloud in Milky Way* *2025*
- UF press release: *UF astronomers use James Webb Space Telescope to uncover hidden stars in the Milky Way's largest stellar nursery* *2025*

Publications (full list: ADS library)

First Author

1. **Nazar Budaiev**, A. Ginsburg, A. T. Barnes, D. Jeff et al., *"JWST's first view of the most vigorously star-forming cloud in the Galactic center — Sagittarius B2."* Accepted to AJ arXiv:2509.11771 — *2026*
2. **Nazar Budaiev**, A. Ginsburg, C. Goddi, Á. Sánchez-Monge, A. Schmiedeke et al., *"Properties of H_2O masers and their associated sources in Sagittarius B2."* ApJ 989, 52 — *2025*
3. **Nazar Budaiev**, A. Ginsburg, D. Jeff, C. Goddi, F. Meng, Á. Sánchez-Monge et al., *"Protostellar cores in Sagittarius B2 N and M."* ApJ 961, 4 — *2024*

Significant Contribution

4. T. Yoo, A. Ginsburg, **Nazar Budaiev**, R. Galván-Madrid, A. Dawson et al., *"A JWST NIRCam/MIRI view of the W51A high-mass star-forming region."* AJ 171, 208 — *2026*
5. T. Yoo, A. Ginsburg, J. Braine, **Nazar Budaiev**, F. Louvet et al., *"ALMA-IMF XX: Core fragmentation in the W51 high-mass star-forming region."* ApJ 994, 233 — *2025*
6. S. Zhang, X. Lu, A. Ginsburg, **Nazar Budaiev**, Y. Cheng et al., *"Subclustering and star formation efficiency in three protoclusters in the Central Molecular Zone."* ApJ 982, L10 — *2025*

Contributing Author

7. A. Ginsburg, S. R. Gramze, M. L. N. Ashby, B. A. L. Gaches, **Nazar Budaiev** et al., *"The colors of ices: Measuring ice column density through photometry."* In review, OJAp arXiv:2510.00292 — *2025*
8. S. Gramze, A. Ginsburg, **Nazar Budaiev**, A. Bulatek, T. Richardson et al., *"Mapping CO ice in a star-forming filament in the 3 kpc Arm with JWST."* In review, OJAp arXiv:2509.21763 — *2025*
9. F. Xu, X. Lu, K. Wang, H. B. Liu (incl. **Nazar Budaiev**), *"DUET: Cloud-wide census of continuum sources showing low spectral indices."* A&A 697, A164 — *2025*
10. A. Ginsburg, J. Bally, A. T. Barnes, C. Battersby, **Nazar Budaiev**, et al., *"A broad line-width, compact, millimeter-bright molecular emission line source near the Galactic center."* ApJ 968, L11 — *2024*
11. D. Jeff, A. Ginsburg, A. Bulatek, **Nazar Budaiev**, Á. Sánchez-Monge et al., *"Thermal properties of the hot core population in Sagittarius B2 Deep South."* ApJ 962, 48 — *2024*
12. Y. T. Yan, C. Henkel, K. M. Menten, Y. Gong, H. Nguyen (incl. **Nazar Budaiev**), *"Discovery of non-metastable ammonia masers in Sagittarius B2."* A&A 666, L15 — *2022*
13. F. Meng, Á. Sánchez-Monge, P. Schilke, A. Ginsburg, C. DePree (incl. **Nazar Budaiev**), *"The physical and chemical structure of Sagittarius B2. VI. UCHII regions."* A&A 666, A31 — *2022*

Contributing Author — ACES Large Program (incl. data reduction)

14. S. N. Longmore, J. Bally, A. T. Barnes, C. Battersby, L. Colzi, A. Ginsburg (incl. **Nazar Budaiev**), *"ACES I: Overview paper."* Accepted to MNRAS arXiv:2602.20340 — *2025*
15. A. Ginsburg, D. L. Walker, Á. Sánchez-Monge, A. T. Barnes, X. Lu, J. E. Pineda (incl. **Nazar Budaiev**), *"ACES II: 3mm continuum images."* Accepted to MNRAS arXiv:2602.20240 — *2025*
16. D. L. Walker, A. Ginsburg, A. T. Barnes, J. Armijos-Abendaño, **Nazar Budaiev** et al., *"ACES III: Molecular line data reduction and HNCO & HCO^+ data preview."* Accepted to MNRAS arXiv:2602.20276 — *2025*
17. X. Lu, D. L. Walker, A. Ginsburg, A. T. Barnes, J. Armijos-Abendaño, **Nazar Budaiev** et al., *"ACES IV: Data of the two intermediate-width spectral windows."* Accepted to MNRAS arXiv:2602.20445 — *2025*
18. P.-Y. Hsieh, D. L. Walker, A. Ginsburg, A. T. Barnes, J. Armijos-Abendaño, **Nazar Budaiev** et al., *"ACES V: CS(2–1), SO, HC_3N, and H40α lines data preview."* Accepted to MNRAS arXiv:2603.00863 — *2025*